Xiao Fu

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Education

Zhejiang University Hangzhou, China
Bachelor in Information Engineering | College of Information Science & Electronic Engineering *Sept. 2018 – Jul. 2022*

- **GPA:** 3.96/4.00, 90.53/100, **Ranking:** 4/145
- **National Scholarship awardee, Twice**
- **Courses:** Computer Vision(96),Digital Signal Processing(95),Matrix Theory(94),Stochastic Process(96),Probability and Mathematical Statistics(95),Numerical Analysis(92),Signals and Systems(98),Calculus(A)(90)

Publications

∗ **indicates equal contribution**

- [2] Xiao Fu*, Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao, "**Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**," Under Review of *International Conference on 3D Vision (3DV)* 2022 [**Paper**|**Project**|**Code**]
- [1] Guangyi Zhang*, Xiao Fu*, Qiyu Hu, Yunlong Cai, Guanding Yu, "**Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network**," *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 [**Paper**|**Code**|**Slides**]

Research Experience

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer Max Planck Institute for Intelligent Systems
Supervisor: Prof. Andreas Geiger and Prof. Yiyi Liao *Sept. 2021 – Mar. 2022*

- Propose Panoptic NeRF, which aims for obtaining per-pixel 2D semantic and instance labels from easy-to-obtain coarse 3D bounding primitives. Our method utilizes NeRF as a differentiable tool to unify coarse 3D annotations and 2D semantic cues transferred from existing datasets. This combination allows for improved geometry guided by semantic information which enables rendering of accurate semantic maps across multiple view. By inferring in 3D space and rendering to 2D labels, our 2D semantic and instance labels are multi-view consistent by design

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network Zhejiang University
Supervisor: Prof. Guanding Yu *Jun. 2020 – Feb. 2021*

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network (DLDUNN) which unfolds the iterative Penality Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

Honors and Scholarships [Certificate & Report]

- National Scholarship **(Highest Honor for Chinese Undergraduates, Twice)** *2020,2021*
- ISEE Honor Student Award **(Top 5 students among ISEE 314 undergraduates)** *2021*
- ISEE Yuelun Alumni Scholarship **(Top 10 teams out of ISEE undergraduates and graduates)** *2020,2021*
- First-Class Scholarship for Academic Excellence of Zhejiang Univeristy**(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 3%)** *2019*
- Outstanding Graduation Thesis of Zhejiang University *2022*
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 7% out of 26,112 teams)** *2021*
- Gold Prize – 7^{th} International "Internet+" Innovation and Entrepreneurship Contest **(Team Leader)** *2021*
- Gold Prize – 12^{th} "Challenge Cup" National College Student Business Plan Competition *2020*
- Grand Prize – "Yongdian Cup" Innovation and Entrepreneurship Contest **(Team Leader)** *2019*
- Grand Prize – "Midea" Global Technology Innovation Contest *2020*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*
- Zhejiang University Five-Star Volunteer *2021*

浙江大學 信息与电子工程学院
College of Information Science & Electronic Engineering, Zhejiang University

CERTIFICATE

This is to certify that Mr Xiao Fu ranks the 4th among the 145 students with the specialty of Information Engineering.

College of Information Science & Electronic Engineering

Zhejiang University

28/09, 2021

浙江大学 信息与电子工程学院

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Zhejiang University
Student's Academic Records

Name: FU Xiao	College/Dept.: College of Information Science and Electronic Engineering		Speciality: Information Engineering	Student ID: 3180101153
Sex: Male	Birthday: 07/19/2000	Birth Place: Sichuan	Entrance Date: 09/01/2018	Graduation Date: 06/30/2022	Years of Program: 4Years

Academic Year 2018-2019			Electronic Engineering Training（A）	1.5	90	Economics	2.0	95	IT Engineering Ethics and Project Management	1.5	92
Courses(1st Term)	***Cr**	***Sc**	Engineering Graphics	2.5	70	Stochastic Process	1.5	96	Introduction to Data Mining	2.0	84
Introduction to Mechanics	1.5	A+	**Academic Year 2019-2020**			Digital System Design	4.0	91	Principles of Communications	3.0	89
The Concise History of Western Music	1.5	93	**Courses(1st Term)**	***Cr**	***Sc**	Discrete Mathematics	2.5	92	**Academic Year 2021-2022**		
Military Training	2.0	89	Complex Variable Functions & Integral Transformation	1.5	86	**Academic Year 2020-2021**			**Courses(1st Term)**	***Cr**	***Sc**
Introduction to Modern Bionics	1.5	89	Software Technology	2.0	88	**Courses(1st Term)**	***Cr**	***Sc**	Project Internship	2.0	A
Mental Education and Foundation of Law	3.0	90	Understanding Internship	2.0	99	Design Experiment of Intelligent Mobile System	2.0	100	iBT-Reading	1.5	93
College English Band III	3.0	84	Electronic Circuit Design Experiments Ⅰ	0.5	83	Matrix Theory	2.0	94	Computer Vision	2.0	96
Calculus（A）Ⅰ	5.0	87	University Physics (A) Ⅱ	4.0	89	Career Planning B	1.5	96	**Courses(2nd Term)**	***Cr**	***Sc**
Wireless Network Application	1.5	99	Military Theory	1.5	97	English ProficiencyTest○	1.0	P	Situation and Policy Ⅱ	1.0	P
Line Dance	1.0	92	Basics of Electronics and Circuits	5.0	89	Business Model	1.0	87	Final Term Project	10.0	92
Linear Algebra（A）	3.5	88	Taekwondo Training	1.0	94	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	2.0	84	Physical test○	1.0	P
Introduction to Information Science and Electronic Engineering	2.0	89	Numerical Analysis	2.0	92	Intro.to Mao Thought & Theoretical System of China Socialism	5.0	90	Classroom Ⅱ: On-campus Practice Activities○	4.0	A
Fundamentals of C Programming	3.0	88	Engineering Training	1.5	85	Digital Signal Processing	3.0	95	Classroom Ⅲ. Out-of-campus (Domestic) Practice Activities○	2.0	A
Courses(2nd Term)	***Cr**	***Sc**	College Physics Experiment	1.5	88	Data Analysis and Algorithm Design	3.0	89			
Introduction to Engineering Ethics(A)	1.5	97	Probability and Mathematical Statistics	2.5	95	Information, Control & Computing	3.0	97			
Ordinary Differential Equations	1.0	86	**Courses(2nd Term)**	***Cr**	***Sc**	Classroom Ⅳ: Exchange Programs in Hong Kong, Macao, Taiwan and foreign countries○	2.0	P			
Situation and Policy Ⅰ	1.0	85	Team Communication & Leadership	2.0	85	**Courses(2nd Term)**	***Cr**	***Sc**			
College English Ⅳ	3.0	88	Optimization Algorithms	3.0	quit	Principles of Embedded Computing System Design	2.0	93			
Calculus（A）Ⅱ	5.0	90	Introduction to the Principle of Marxism	3.0	95	Scientific Research Training Program II	2.0	quit			
Study on The Book of Master Zhuang	3.0	95	Signals and Systems	4.0	98	Photoelectric information processing comprehensive experiment	2.0	90			
Lectures on Programming	2.0	87	Electromagnetic Fields & Waves	4.0	95	Lab. of Principles of Communication	1.0	93			
University Physics (A) Ⅰ	4.0	88	Swimming (Basic Level)	1.0	98	Artificial Intelligence	3.0	91			
Cross-Country Orienteering (Basic Level)	1.0	95	Electronic Circuit Design Experiments II	1.0	94	Digital Image Processing	3.0	82			
Modern Chinese History	3.0	90	Experiment of Digital System Design	1.0	96	Network and Communication Security	2.0	85			

浙江大学
成绩考核专用章
（1）

Overall GPA:3.95/4.0(90.53/100)

Degree Granted:	Credits Required for Graduation: 157+8+6	Credits Obtained: 176.5

Three grade systems are used simultaneously in Zhejiang University,specifically as follows(*Cr=Credits;*Sc=Score):
1.The percentage system:Above 60 is passing,100 is full mark.
2.Five degree grading:Excellent(A),Good(B),Fair(C),Passing(D),Failed(E).
3.Two degree grading:Passing(P),Failed(F).
4.Courses marked with * are courses transferred from other universities and their original records are kept.
5.Courses marked with "△" are repeat courses, and GPA is calculated on the highest grade.
6.Courses marked with "○" are not included in the GPA calculation.

Dean,Undergraduate School 张光新

Date Issued:06/07/2022

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RECOGNITION INSTRUCTION: 1."ZHEJIANG UNIVERSITY" is microprinted on the lower right corner as a line, they can be seen under the magnifying glass. 2.The fluorescent school badge of ZHEJIANG University on the higher left corner will appear under the UV light. 3.The words "ZJU" on the center of the report will turn purple under the sunlight. 4.This style transcript has been formally in use since September 1,1999.